Triangle (Minimum Path Sum)

HackerRank

You are given an R × R 2D array of integers representing a triangle. The elements of the triangle are located in the lower-left half of the grid (where column index j ≤ row index i). The missing elements in the upper-right half of the grid (where j > i) are all represented by the value −1.

Normally, from a valid element at row i and column j, you can step down to row i + 1 at column j or column j + 1.

However, you are granted a special ability: you may jump over exactly one row during your descent. This means from index j on row i, you can choose to skip row i + 1 entirely and land on row i + 2 at index j, j + 1, or j + 2. You may use this jump at most once on your path.

Find the minimum possible path sum from the top element (row 0, column 0) to any valid element in the bottom row (row R − 1).

Input Format

The first line contains a integer, R, denoting the dimensions of the R x R grid.

Each of the R lines contains R space-separated integers, representing row i of grid.

Constraints

1 <= R <= 200

-10^4 <= grid[i][j] <= 10^4

Output Format

The output contains an integer value

Sample Input 0

```
3
1 -1 -1
10 10 -1
1 1 1
```

Sample Output 0

```
2
```

Explanation 0

Without the jump, you would have to pass through a 10 on row 1, resulting in a minimum path sum of 1+10+1 = 12. However, from the top (1), you can jump over row 1 directly to one of the 1's on row 2. The minimum path sum is 1 + 1 = 2.

Sample Input 1

```
2
6 -1
-2 -3
```

Sample Output 1

```
3
```

Explanation 1

Since there are only 2 rows, jumping is impossible as you would land outside the valid triangle. You must take a normal step. Moving from 6 to −3 gives the minimum sum of 6 + (−3) = 3.

Sample Input 2

```
4
0 -1 -1 -1
5 5 -1 -1
-10 10 10 -1
5 5 -10 5
```

Sample Output 2

```
-5
```

Explanation 2

From the top (0), you can move down to the right 5 (row 1, index 1). From there, you use your jump over row 2 directly to the −10 on row 3 (index 2). The path sum is 0 + 5 + (−10) = −5. Alternatively, jumping over row 1 (index 2) to the −10 on row 2, then stepping to the right 5 on row 3 also yields −5. The minimum sum is −5.